swarm deployment and intelligent traffic management in urban airspaces. Additionally, quantum navigation systems and blockchain-based communication security are being explored to enhance reliability and prevent cyber threats in autonomous operations.

Applications of fifth-generation drones extend across defense, logistics, smart cities, precision agriculture, and environmental sustainability. Autonomous delivery systems, infrastructure monitoring swarms, and defense-oriented reconnaissance units exemplify how intelligent UAVs are reshaping global industries. Governments and the private sectors are now focusing on integrating autonomous aerial ecosystems with regulatory frameworks that ensure safety, ethics, and airspace management.

Figure 1.7 illustrates a quadcopter-based fourth-generation drone equipped with multi-sensor modules, GPS, LiDAR, and onboard cameras. The figure highlights the integration of a flight controller and an AI-assisted navigation system, enabling semi-autonomous operation. Various application domains, such as agriculture, construction, environmental monitoring, and logistics, are represented through icons surrounding the drone, showcasing its multifunctional commercial role and technological versatility.